Task 2: Login and Registration System

CodeAlpha C++ Internship | Marathi + English explanation

Program kay karto? (What it does)

• User la register karto (username + password) ani nantar login verify karto.
• Credentials users.txt file madhe save hotat - password plain text nahi, salt + hash karun.

Features

• Username validation: 3-15 characters, fakt letters, digits ani underscore.
• Password validation: kamit kami 6 characters, letters + digits donhi pahije.
• Duplicate username check: same username parat register hot nahi.
• Success / error messages pratyek step la.

Code kasa kam karto? (How the code works)

• makeSalt(): random 8 characters cha salt banavto, pratyek user cha vegla.
• hashPassword(): salt + password cha hash banavto (std::hash). File madhe asli password kadhi save hot nahi.
• registerUser(): validate -> duplicate check -> hash -> file madhe append (ios::app).
• loginUser(): file vachto, username shodhto, entered password cha hash compare karto.
• Format: username salt hash (pratyek line = ek user).

Important note

• std::hash he learning purpose sathi aahe. Real apps madhe bcrypt / argon2 vapra. Video madhe he nakki sanga - tyane tumcha understanding disel.

Video madhe kay bolaych (Video tips)

• Register kara, mag users.txt ughadun dakhava ki password hashed aahe.
• Same username parat try kara (error), chukicha password try kara (error), shevti correct login.

Kasa run karaycha (How to Run)

```
g++ -std=c++17 CodeAlpha_Login_Registration_System.cpp -o login
./login            (Windows: login.exe)
```

Full C++ Code

```
// CodeAlpha C++ Internship - Task 2: Login and Registration System
// NOTE: Passwords are salted + hashed (educational). For real apps use bcrypt/argon2.
#include <iostream>
#include <fstream>
#include <string>
#include <sstream>
#include <functional>
#include <cctype>
#include <cstdlib>
#include <ctime>
using namespace std;

const string USER_FILE = "users.txt";   // format: username salt hash

// ---------- Helper functions ----------
string makeSalt() {
    const string chars = "abcdefghijklmnopqrstuvwxyz0123456789";
    string s;
    for (int i = 0; i < 8; i++) s += chars[rand() % chars.size()];
    return s;
}

string hashPassword(const string &password, const string &salt) {
    size_t h = hash<string>{}(salt + password);
    stringstream ss;
    ss << h;
    return ss.str();
}

bool validUsername(const string &u) {
    if (u.length() < 3 || u.length() > 15) return false;
    for (char c : u)
        if (!isalnum((unsigned char)c) && c != '_') return false;
    return true;
}

bool validPassword(const string &p) {
    if (p.length() < 6) return false;
    bool digit = false, alpha = false;
    for (char c : p) {
        if (isdigit((unsigned char)c)) digit = true;
        if (isalpha((unsigned char)c)) alpha = true;
    }
    return digit && alpha;
}

bool usernameExists(const string &username) {
    ifstream in(USER_FILE);
    string u, salt, h;
    while (in >> u >> salt >> h)
        if (u == username) return true;
    return false;
}

// ---------- Registration ----------
void registerUser() {
    string username, password;
    cout << "\n--- REGISTRATION ---\n";
    cout << "Username (3-15 letters/digits/_): ";
    cin >> username;

    if (!validUsername(username)) {
        cout << "Error: Invalid username format.\n";
        return;
    }
    if (usernameExists(username)) {
        cout << "Error: Username already taken. Try another.\n";
        return;
    }

    cout << "Password (min 6 chars, letters + digits): ";
    cin >> password;
    if (!validPassword(password)) {
        cout << "Error: Weak password. Use at least 6 characters with letters and digits.\n";
        return;
    }
    string salt = makeSalt();
```

```
    ofstream out(USER_FILE, ios::app);
    if (!out) {
        cout << "Error: Could not open file.\n";
        return;
    }
    out << username << " " << salt << " " << hashPassword(password, salt) << "\n";
    cout << "Success: Registration completed! You can now login.\n";
}

// ---------- Login ----------
void loginUser() {
    string username, password;
    cout << "\n--- LOGIN ---\n";
    cout << "Username: ";
    cin >> username;
    cout << "Password: ";
    cin >> password;

    ifstream in(USER_FILE);
    if (!in) {
        cout << "Error: No users registered yet.\n";
        return;
    }
    string u, salt, h;
    while (in >> u >> salt >> h) {
        if (u == username) {
            if (hashPassword(password, salt) == h)
                cout << "Success: Login successful! Welcome, " << username << ".\n";
            else
                cout << "Error: Incorrect password.\n";
            return;
        }
    }
    cout << "Error: Username not found.\n";
}

int main() {
    srand((unsigned)time(0));
    int choice;
    do {
        cout << "\n=========================\n";
        cout << " LOGIN & REGISTRATION\n";
        cout << "=========================\n";
        cout << "1. Register\n2. Login\n3. Exit\nChoice: ";
        if (!(cin >> choice)) {
            cin.clear();
            cin.ignore(1000, '\n');
            choice = 0;
        }
        switch (choice) {
            case 1: registerUser(); break;
            case 2: loginUser();    break;
            case 3: cout << "Goodbye!\n"; break;
            default: cout << "Invalid choice. Try again.\n";
        }
    } while (choice != 3);
    return 0;
}
```

Sample Output

```
=== Menu: 1 Register, 2 Login, 3 Exit ===

Register rahul_1 / abc123  -> Success: Registration completed! You can now login.
Register rahul_1 again     -> Error: Username already taken. Try another.
Login rahul_1 / abc123     -> Success: Login successful! Welcome, rahul_1.
Login rahul_1 / wrong1     -> Error: Incorrect password.

users.txt content:
rahul_1 0mdi9z3r 9167638685881775932
```